PHISHING URL DETECTION SYSTEM USING ENSEMBLE LEARNING

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ABSTRACT

Recent years, phishing attacks have become one of the most substantial cybersecurity threats that cybercriminals aim to deceive users into giving out their sensitive information like usernames, passwords, and bank details through fake URLs or websites. As the techniques used in phishing attacks keep on evolving, traditional methods of detection are often stumped, which is one of the reasons why traditional detection algorithms fail to detect the newly created phishing URLs. The main of this Final Year Project is to create a phishing URL detection system using Ensemble Learning to improve the accuracy and reliability of the system. Furthermore, the proposed system processes the features based on URLs and uses the various machine learning models, including Random Forest, Gradient Boosting, and XGBoost, to classify URLs as either phishing or legitimate. These models are combined with an ensemble model to enhance the overall performance. The main parts of the development are data collection, data preprocessing, feature extraction, model training, and performance evaluation. Besides, the system is evaluated with the use of standard metrics such as accuracy, precision, recall, and F1-score. Experimental results show that the ensemble learning model can outperform the individual models and provide more accurate and consistent detection results. A simple and user-friendly interface is also developed to allow the users to input URLs and efficiently get the classification results. This project demonstrates that ensemble learning is a good and effective solution to phishing URL detection and will help in reducing the risk of online fraud and raising user awareness about cybersecurity.

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LIST OF ABBREVIATIONS/ SYMBOLS

API	Application Programming Interface
APWG	Anti-Phishing Working Group
CPU	Central Processing Unit
DNS	Domain Name System
DT	Decision Tree
FIST	Faculty of Information Science & Technology
FYP	Final Year Project
GB	Gradient Boosting
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
ML	Machine Learning
MMU	Multimedia University
NB	Naïve Bayes
NPU	Neural Processing Unit
OS	Operating System
QR	Quick Response
QRLJacking	Quick Response Code Login Jacking
RF	Random Forest
SDLC	Software Development Life Cycle
SEG	Secure Email Gateway
SMS	Short Message Service
SSL	Secure Sockets Layer
SVM	Support Vector Machine
UI	User Interface
URL	Uniform Resource Locator
XGBoost	Extreme Gradient Boosting

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CHAPTER 1

INTRODUCTION

1.1 Overview

In today's digital era, the Internet plays a vital role in communication, education, business, and entertainment. However, this increasing dependence on online services has also made users vulnerable to various cybersecurity threats, with phishing being one of the most common and dangerous. Phishing is a trick where attackers pretend to be legitimate organizations to steal sensitive information from users, such as passwords, banking details, or personal credentials. These attacks often happen through fake websites, fraudulent emails, or harmful QR codes that lead victims to imitation pages.

Besides, old Phishing prevention methods or tools, like blacklist and rule-based filters, are no longer helpful against nowadays advanced phishing tactics. Attackers frequently change URLs using domain hiding, URL shortening, and fast-flux hosting to avoid detection. As a result, we need a smart and flexible detection system to quickly show the difference between real and fake URLs.

Moreover, Machine Learning (ML) has shown to be a useful method for phishing detection, as it can find hidden patterns in large data sets. Particularly, ensemble learning, which combines several classifiers to boost prediction accuracy, has shown better results that single-model methods. This Final Year Project proposes creating a Phishing URL Detection System using ensemble learning, which will analyse URL features to identify phishing websites. The system will also have a QR code scanning feature to spot phishing links in QR codes. The application will be developed with Flutter, allowing for a user-friendly experience on both Android and iOS devices.

1.2 Problem Statement

Phishing attack are still a major threat to online users by taking advantage of human trust and the weaknesses in technology. Cybercriminals often set up fake URLs and websites to trick people into giving their personal or financial information. Despite ongoing efforts in cybersecurity education, many users, particularly older smartphone users with limited technology knowledge, remain at risk. With the rise of mobile devices and the use of QR codes in payment, menus, and ads have increased this risk, as users often scan links without checking if they are safe.

The desired future state is an online environment where users can confidently access websites and scan QR codes without worrying about the hidden threats. To achieve this, we need a smart, effective, and user-friendly detection system that can quickly check whether a URL or QR code leads to a legitimate or phishing website. However, current solutions do not meet this need. Traditional blacklist or rule-based systems cannot identify newly created phishing URLs, and most of the existing machine learning models use single classifiers that have difficulty keeping up with changing phishing tactics or perform inconsistently across different datasets. Furthermore, many tolls are not designed for mobile devices, which limits access for everyday users.

On top of that, this gap shows a clear need for a more flexible and mobile friendly phishing detection system. By using ensemble learning, which combines multiple classifiers to improve detection accuracy, the proposed Phishing URL Detection System using Ensemble Learning aims to fill this gap. This system will be available as a cross platform mobile application. It will let users manually paste the URLs or scan QR codes to detect phishing attempts in real time. Tackling this issue not only improves cybersecurity but also helps users, especially those with limited technical knowledge, to let them browse safely in a more digital world.

1.3 Project Objectives

The main goal of this Final Year Project (FYP) is to design and develop a user-friendly mobile based phishing detection system that uses ensemble learning techniques to identify and prevent phishing attacks from the cybercriminals. The project aims to detect phishing attempts from both manually pasted URLs and QR codes, which attackers increasingly exploit. To achieve this goal, the project follows these detailed objectives:

- To study current existing phishing detection techniques in terms of their strengths and limitations of traditional detection methods in phishing such as blacklist-based detection, filters based on rules, and machine learning methods based on a single model.
- To identify and examining the necessary URL-based features that can very well differentiate between the real and the fraud websites, highlighting the feature selection and preprocessing strategies.
- To create a model for phishing URL detection based on ensemble learning using classifiers like Random Forest, Gradient Boosting, and XGBoost, but rather than full implementation focusing on model architecture and integration.
- To design the architecture of a cross-platform mobile application that would be built using Flutter and supports manual URL input and QR code scanning for phishing detection.
- To plan suitable evaluation metrics such as accuracy, precision, recall, and F1-score for measuring the performance of the system in the next phase of the project.
- To propose potential enhancements and future improvements for phishing detection systems based on the findings of the study.

By meeting these objectives, the project aims to create a valuable phishing detection solution that contributes to the fields of cybersecurity and machine learning while also providing a practical tool to help users protect themselves from phishing threats on mobile platforms.

1.4 Project Scope

The scope of this Final Year Project sets the limits for the Phishing URL Detection System using Ensemble Learning. It ensures the project stays focused, achievable, and fits within the available time, resources, and technical limits. The main goal is to create a mobile application that can analyse URLs and QR codes to detect phishing attempts. This project combines machine learning techniques with user-friendly mobile features. This system aims to help general users, especially older smartphone users who may not be aware of cybersecurity risks, by providing an easy and reliable wat to check suspicious links.

Moreover, the project includes several key tasks and deliverables. It will involve gathering publicity available datasets of phishing and legitimate URLs, preprocessing these datasets, and extracting important URL-based features for training the model. The system will use ensemble learning models, such as Random Forest, Gradient Boosting, and XGBoost, to improve detection accuracy and reliability. A cross-platform mobile application will be developed using Flutter. It will feature both manual URL input and QR code scanning for real-time phishing detection. The project also evaluates the system's performance using standard metric like accuracy, precision, recall, and F1-score.

At the same time, the scope clearly defines what is not included. The project does not plan to integrate large-scale real-time threat intelligence or perform deep webpage content analysis. It will not deploy the system as a commercial product. Advanced security features like browser extensions, authentication systems, or enterprise-level backend infrastructure are also outside the project scope. The focus is solely on developing a functional prototype that shows how ensemble learning can be used for phishing detection in a mobile setting.

Lastly, by outlining what is included and excluded, this project scope provides a clear plan and keeps the project realistic and manageable. It helps maintain focus on important goals while delivering a practical and effective phishing detection tool within the academic timeframe.

1.5 Report Organisation

This Final Year Project report is divided into five chapters which are structured to describe the research and planning process in a clear and logical manner. Chapter 1 presents an introduction to the project by giving the background of the study, problem statement, project objectives, project scope, and the overall motivation behind the development of Phishing URL Detection System using Ensemble Learning techniques.

Chapter 2 introduces the literature review where the existing research works related to phishing attacks, methods of phishing attack detection based on URLs, ensemble learning algorithms, phishing threats using QR codes, and mobile security applications are discussed. This chapter highlights the limitations of the current approaches and the research gaps that support the need for the proposed system.

In Chapter 3, this chapter provides the proposed methodology and system design. It includes the processes for dataset selection, feature extraction, ensemble learning model planning, system architecture design, and mobile application development strategy. Besides, this chapter presents the project implementation from the machine learning and mobile application perspectives.

Chapter 4 deals with the overall project planning and discussion which is applicable to the Final Year Project Phase 1. It contains system workflow explanations, development tools, and implementation considerations that get the project ready for the next phase.

Lastly, Chapter 5 wraps up the report by giving a summary of the work that has been done in Final Year Project Phase 1, by review what has been achieved in fulfilling the objectives of the project and outline the possible directions for future development in Final Year project Phase 2.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

The 21st-century digital ecosystem is experiencing a significant change, shifting from a static, desktop-focused computing setting to a dynamic, mobile-first environment. During such change, the corresponding development of cyber threats has been witnesses, and among them, phishing attacks have become the most prevalent and complex means of targeting information security. Phishing, which is a type of semantic attack based on social engineering, the objective of which is to deceive the user into revealing sensitive credentials, has transitioned from the simple method of email spoofing to multi-channel campaigns employing SMS (”Smishing”), social media, and QR code (“Quishing”) among others. Not only has the number of incidents where mobile phones were used as the point of phishing attack grown, but it has also been indicated by APWG (Anti-Phishing Working Group), and cybersecurity reports published in 2024 that this is because built-in UI limitations on smartphones, such as cut-off URLs or hidden address bars, conceal normal indicators of being deceived (Alkhalil et al., 2021). This chapter provides on the entire and critical analysis of the existing phishing detection systems as a means of theoretical and empirical justification for the “Phishing URL Detection System using Ensemble Learning” proposed. The systematic research mainly comprised publications from 2022 – 2025, has been called upon to critically evaluate the performance of the methods of Ensemble Learning such as Random Forest, Gradient Boosting, and XGBoost on resource-limited Android mobile devices. Additionally, it addresses the emerging threat of QR-based phishing and exposes weaknesses in defence systems to thereby demonstrate the need for a detection paradigm that is unified, user-friendly, and capable of real-time operation.

2.2 Theoretical Foundations of Phishing Detection

Building a strong phishing detection system demands first to reveal how the modern phishing works and the weaknesses of past defence strategies. Traditional defence methods mainly relied on list-based approaches, which kept static databases of allowed domains (allow-list) and known malicious domains (deny-lists). While these methods are computationally efficient, research consistently points out the "window of vulnerability" in this reactive model. Recent studies show that the average lifespan of a modern phishing URL is often less than 24 hours. Many "zero-day" attacks disappear before they can be added to blacklist services like Phish Tank or Open Phish (Oest et al., 2020). For this reason, experts now favour machine learning (ML) methods. These approaches view phishing detection not as a simple signature-matching task, but as a predictive classification problem that can identify malicious intent based on hidden patterns and structural oddities.

The success of any ML-based system depends on the quality of feature engineering, which involves extracting measurable attributes from raw URL data. Research divides these features into three main categories, each with different effects on mobile app development. Lexical features come solely from the text of the URL. They include statistical aspects such as URL length, the number of special characters like dots, hyphens, and '@', and character distribution entropy. Recent studies by (Haynes et al., 2021) and updates in 2024 highlight that lexical features are lightweight and protect user privacy because they do not require the device to connect to a malicious server, thus preventing drive-by downloads.

Moreover, host-based features involve checking external services to gather contextual metadata, such as the domain's registration age, DNS record validity, and SSL certificate issuer. While they are powerful, research by (Rao et al., 2025) points out that excessive reliance on host-based features can create network delays, harming the user experience. Content-based features involve analysing the HTML source code and JavaScript of the target page. They provide the most context but are resource-intensive and carry significant security risks by requiring the device to fetch potentially harmful content. Therefore, for a mobile-based FYP, the literature strongly advocated for a mixed feature engineering approach that focuses on detailed lexical analysis

supported by light host-based checks to balance detection accuracy with application performance.

2.3 Ensemble Learning Classifier: A Comparative Analysis

The classifiers that are most employed individually for the purpose of phishing detection include DT (Decision Trees), SVM (Support Vector Machines), and NB (naïve bayes). However, these models are prone to bias and variance issues, which at high variance rates result in overfitting, and at high bias rates, to underfitting. It is especially hard to cope with such issues when one deals with the complex and noisy datasets that are typical of Phishing URLs. Therefore, the recent research is directed toward Ensemble Learning, which is a new trend in combination of several 'weak' learners to produce a single 'strong' one processing better generalisation capabilities. The three algorithms at the core of this study are Random Forest, Gradient Boosting, and XGBoost.

2.3.1 Random Forest (RF): Stability via Bagging

RF is considered as one of the "Bagging" (Bootstrap Aggregating) methods for classification. The procedure includes creating large number of decision trees during training and then the one with the most votes from the individual trees is chosen as the output. A very important advantages of RF, which author EI Attaoui refers to in the year 2024, is the model's ability to resist overfitting. the concept of training each tree on a randomly chosen data sample (bootstrap sample) along with utilizing only a random subset of features for each tree results in a collection of trees that are inherently diverse, with errors that are uncorrelated and thus averaged out. In such comparative studies, RF continues to demonstrate accuracy rates of about 96% to 97% which are very high, and it also delivers "Feature Importance" scores that are valuable for the developers in identifying which URL features for instance the "presence of IP address" are most probably related to phishing. Still, the large size of the RF models might be a problem that calls for mobile optimisation.

2.3.2 Gradient Boosting and XGBoost: Precision via Boosting

The method of tree building in Gradient Boosting (GB) is quite different from Bagging, as the latter relies on the simultaneous creation of trees. In Gradient Boosting, the tree constructing process is done in an orderly manner, where the last tree tries to enhance the first one. XGBoost (eXtreme Gradient Boosting) is the most sophisticated version of this framework that has conquered tabular data classification. In fact, the years 2023 to 2025 studies extensively refer to XGBoost as the standard for phishing detection because of its system optimisation (parallel tree construction) and algorithmic enhancements (L1 and L2 regularization). A pivotal study by (Sugianto et al., 2024) stated that XGBoost attained an accuracy of 97.27% on an extensive URL dataset, which is meanwhile, a tiny bit more than Random Forest (95.66%), but still very far away from traditional Neural Networks. One of the features that make XGBoost a real candidate for the mobile application's real-time inference requirements is its capability to deal with sparse data, which is a common characteristic of feature vectors, and its fast execution.

Table 1 : Comparative Performance of Ensemble Algorithms in Recent Literature (2023 – 2025)

Algorithm	Ensemble Type	Strengths	Weaknesses	Mobile Suitability	Average Accuracy
Random Forest (RF)	Bagging	High stability; resistant to overfitting; interpretable feature importance	Large model size, slower inference than optimized Boosting	High (if pruned)	95.6% - 97.5%
Gradient Boosting (GB)	Boosting	Low bias, high predictive accuracy on complex datasets	Sequential training is slow, prone to overfitting if not tuned	Medium	96.0% - 97.5%
XGBoost	Boosting	Highest accuracy, regularization (L1/L2), handles sparse data, fast inference	Many hyperparameters to tune, complex black-box model	Very High	97.2% - 99.2%

2.4 Mobile-Based Detection Architectures

Implementing a phishing detection system on Android involves specific architectural constraints that are not present in desktop environments. The main points of the literature are: Limited Screen Real Estate, which makes it impossible to display security indicators; Resource Constraints (Battery/CPU), which reduces the number of complexities in on-device models; and OS-Level Sandboxing, which limits the communication of apps with the browser's traffic.

2.4.1 Cloud-Based vs. On-Device Inference

The literature identifies two main ways of architectural design which is Cloud-Based Offloading versus On-Device Inference. The cloud-based systems work by transmitting the examined URL to a distant server for processing. Although this allows for the use of extensive resources and advanced models such as Deep Learning, it also introduces issues like network latency and significant privacy concerns about the user's search history. On the other hand, the On-Device method, which has become the preference of many recent privacy-focused studies, conducts classification on the CPU/NPU of the smartphone. A significant study by (Reo et al., 2025) introduced "Phish-Jam," a mobile-specific ensemble model that reaches 98.93% accuracy with simple features that are completely processed on the device. This confirms that running optimised ensemble models (translated via Tensor Flow Lite) on Android is practical without exhausting the battery or requiring a constant internet connection, which is in total harmony with this project's aims.

Table 2 : Comparison of Mobile Phishing Detection Architectures

Architecture	Processing Location	Latency	Privacy	Offline Capability
Cloud-Based API	Remote Server	High (Network dependent)	Low (Data send to cloud)	No (Fails without internet)
On-Device (Proposed)	Local CPU / NPU	Ultra-low (<100ms)	High (Data stays on device)	Yes
Hybrid	Local + Cloud	Medium	Medium	Partial

2.5 The Rising Threat of QR Code Phishing ("Quishing")

The biggest chance in the phishing world, according to the publications in the time frame of 2024 – 2025, is the rise of QR code phishing, also known as "Quishing" attacks. This method exploits the shift from physical to digital, thus bypassing traditional email security gateways (SEG), which can scan for malicious text but not image-based threats.

2.5.1 The "Black Box" Problem and Quick Response Code Login Jacking ("QRLJacking")

The primary threat of a QR code is that it acts like a black box. Users are not able to verify the destination URL until after scanning the code, at which point the malicious resource will be automatically loaded by the browser. Attackers take advantage of this to launch QRLJacking in which a victim scans a malicious QR code, thinking it is for a legitimate login, for example WhatsApp Web, but actually will be giving the attacker access to the person account.

2.5.2 Virtual vs. Content Analysis

Among the current detection methods, there are two approaches can be pointed out. One consists of payload extraction that includes decoding the QR code to access the URL and subsequently analysing the URL using standard classifiers. Although this method is effective, is still reactionary and always must deal with URL resolutions. A more novel and deeper approach in which visual or structural analysis is being discussed in the "QRïS" framework by (Akram et al., 2025). Their research shows that malicious QR codes typically show clear structural abnormalities, such as high levels of error correction, which is deployed to encode information such as logos or to corrupt the code to avoid automated scanning, and certain data density patterns. By analysing these raw visual features using machine learning, particularly XGBoost, it becomes possible to predict "malicious intent" before processing the URL. This is an important area for research and one that most commercial mobile scanners are lacking.

Table 3 : Methodologies for QR Code Phishing Detection

Methodology	Mechanism	Advantages	Limitations
Payload Analysis (Traditional)	Decode QR ↓ Extract URL ↓ ML Classification	Leverages mature URL detection models	Reactive; requires decoding the payload
Visual Analysis (Novel / QRïS)	Analysing QR image structure, error correction, and pixel density	Pre-emptive; detects manipulation before decoding	Emerging field; requires advanced image processing

2.6 Critical Analysis and Research Gaps

Despite the intensive research performed conducted on phishing detection, there are several critical gaps to be addressed, which the proposed FYP will try to fill. Current studies often limited to a single domain of detection, many studies look at either text-based analysis of URLs or the interpretation of QR codes, as opposed to combining both into one united solution. Very few investigations are described describing a mobile application that is an ensemble learning URL scanner and visual QR code scanner that is in a seamless workflow. Additionally, while previous studies, including those from (Sugianto et al., 2024), showed that models such as XGBoost work very well in desktop environments, there is a lack of literature that explains how computational heavy ensemble models can be compressed or optimized for Android implementation through tools such as Tensor Flow Lite for real-time performance. Another major gap involves feature integration, as top mobile security solutions still rely primarily on static rules or blacklists. Therefore, there is little exploration in systems that combine lexical features of URLs with visual features of QR code to offer a layered and adaptive security, which is still not supported by most existing app-store security applications.

2.7 Summary of Key Insights and Identified Gaps

In summary, the literature reviewed shows that phishing has evolved into a multi-vector threat that is increasingly attacking mobile users using URL and QR-based attacks. Thus, exposing the limitations of traditional blacklist and single-channel protection system. Consistent evidence from research supports the use of machine learning techniques, specifically ensemble techniques like Random Forest, Gradient Boosting and XGBoost, which have shown to be highly effective in handling the complexity and class imbalance of phishing datasets. At the same time, recent studies highlight the need for mobile deployment that require models that are fine-tuned for privacy, latency, and hardware parameters. For instance, the "Phish-Jam" study demonstrates that quantized ensemble models can be efficiently executed on a device through Tensor Flow Lite. Moreover, a large gap in research has occurred as the current solutions rarely provide the combination of lexical analysis of URLs and the visual analysis of a QR code, and this leaves the consumers vulnerable to the growing prevalence of QR code phishing (Quishing) attacks. The lack of coherent detection pipeline makes the proposed FYP relevant, which plans to incorporate ensemble-based URL classification with QR code analysis into a lightweight mobile application. Overall, the findings detailed in this chapter prove the feasibility and novelty of the proposed system and hence, its potential contribution to contemporary mobile cybersecurity.

CHAPTER 3

METHODOLOGY

3.1 Overview

This project uses a Hybrid Software Development Life Cycle (SDLC) methodology which combines the best parts of both the Waterfall and Agile model. The nature of this project that includes both machine learning model development and mobile application prototype creates the need for striking a balance between structured planning and iterative refinement. The Waterfall component offers stability and clarity at the early stage when the requirements, datasets, and system scope need to be well defined. Meanwhile, the Agile component is useful to support the iterative experimentation required for training and optimising ensemble learning models, as well as the step-by enhancement of the mobile application's user interface and system functions. The hybrid methodology makes sure that the system is developed in a structured progression but is flexible enough to include improvements in each iteration. This gives a good foundation to create an accurate, efficient and user-friendly Phishing URL Detection System.

3.2 Hybrid SDLC Model Overview

Hybrid SDLC methodology combines the linear planning robustness of the Waterfall model and the cyclical and feedback-driven nature of Agile development. The development process starts with a Waterfall oriented Planning and Requirements phase, where all project requirements, dataset sources, machine learning model choices and system functional requirements are laid out. once this basic plan is done, the project moves into the Agile cycle which consists of Design, Development, Implementation and Evaluations phases. These phases are sequentially repeated allowing the system to continuously evolve with feedback, testing, and small increments. This approach is particularly beneficial for machine learning development, where the efficiency of the model can only be increased by making a series of changes, and for mobile application development, which traditionally progresses by going through many stages of refinement.

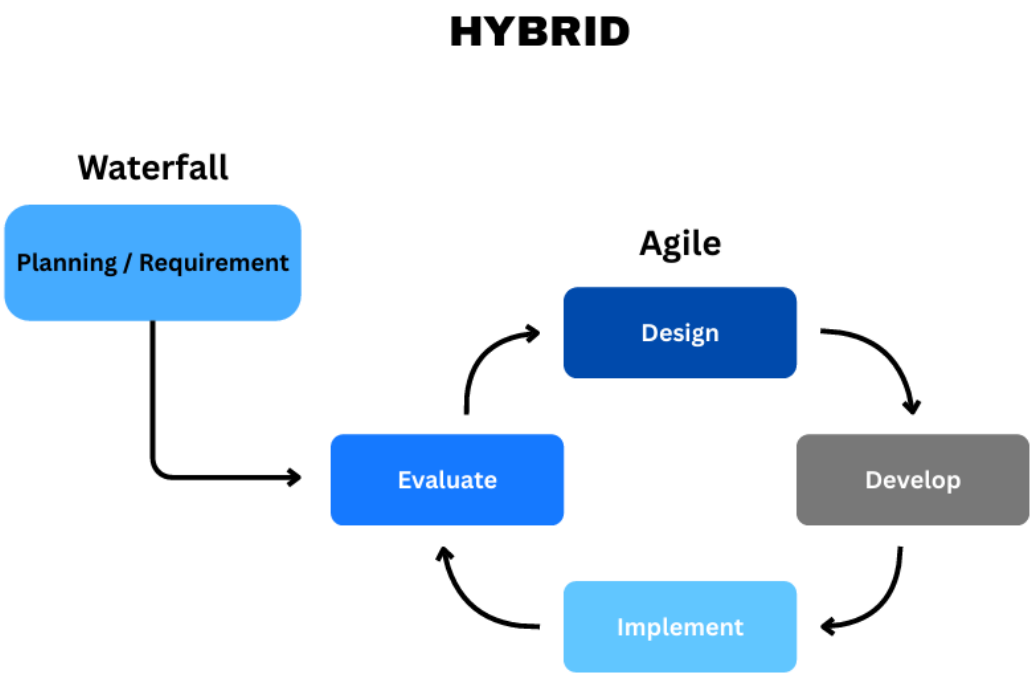

Figure 1: Hybrid SDLC Model for the Proposed System

3.3 Planning and Requirements Phase

The project starts with a full planning phase based on the Waterfall model. During this phase, the entire scope of the Phishing URL Detection System is properly defined and positioned before going into the details. Project requirements are first defined such as system functionalities, the mobile application platform. The machine learning techniques, and the data sources for training. The selection of Random Forest Gradient Boosting, and XGBoost as major classifiers using ensemble learning takes a solid shape in this phase because ensemble classifiers are proven to be more effective at predicting the data compared to single model classifiers for highly varying and adversarial datasets like phishing URLs.

Furthermore, reliable datasets containing labelled phishing and legitimate URLs are identified in the Planning phase. Public repositories such as Phish Tank, Open Phish, and curated Kaggle datasets are chosen for their authenticity and consistency. The scope of features to be extracted from URLs, which includes lexical, domain-based and heuristic attributes are also determined in advance to ensure that model is trained on meaningful indicators.

On the system side, the requirements for the Flutter mobile application are described, including the support of the manual URL input and the QR code scanning to improve usability, enabling to detect phishing URLs from printed materials or other screens. The desired outputs, including binary classification, risk explanation and visual indicators are also defined. Following this structured planning will ensure that not only the machine learning development, but also the mobile app developments is carried out with specific goals and deliverables.

3.4 Design Phase

In Design Phase the phase represents the point where the construction of the system is changed from a structured approach to an iterative approach. In this phase, the conceptual ideas and requirements that were identified during the previous phase are translated into clear architectural blueprints. Because the project combines a machine learning inference engine and a mobile application interface, the design activities are focused on designing a modular architecture that allows each of the components to evolve independently using Agile iterations. The objective for this phase is to determine a well-defined system structure, flow and interaction model, so that both mobile and backend components can be developed seamlessly.

At the highest level, the system is designed in a multi-layer architecture, which consists of the Flutter mobile application, the communication layer, the machine learning inference engine, the output handler, and the result display module. This modular breakdown makes sure that updates to the machine learning model, preprocessing logic or mobile interface, will not require much rework across other components. Modular design is especially critical in phishing detection systems, where new datasets or features and threat patterns can require new iterations and retraining at rapid rates. By logically separating the system, the architecture facilitates the support of long-term extensibility, maintainability, and iterative improvements of the system.

The first component designed was the flutter mobile application which is the interface layer of the system that the user interacts with. Its key functions are collecting input from the user such as manually entered URL or scanned QR code, URL format checking, and interacting with the backend inference engine or the on-device model. The mobile UI (User Interface) design is based on the Material Design principles to provide a clean, intuitive, and accessible user experience. The interfaces that are planned for this phase are homepage with two main options which is “Manually Enter URL” and “Scan QR Code”, QR scanning interface using device’s camera and results page with classification outcomes and explanatory details. Designing these components early means that the system will be user centred.

The second major design activity relates to the communication layer which acts as a bridge between the mobile application and inference engine. If a backend API is being used, this layer will be based on the use of an HTTPS POST request, a JSON payload and lightweight routing mechanisms. If the system is deployed locally on a device, this layer is reduced to an internal function call that deals with processing and inference. No matter what the method of deployment is, this layer is responsible for ensuring that the input data is properly formatted and sent for model evaluation.

The core of the architecture is the backend machine learning inference engine that was designed as an all-in-one pipeline with three sub-models included preprocessing, feature extraction and ensemble learning inference. The preprocessing module is used to normalise the raw URL and removes unwanted characters and check the structure of the URL before extracting features from it. It is the feature extraction engine that turns the URL into numerical feature vector by calculating lexical, domain-based and heuristic attributes. These extracted features are perfectly aligned with the features used during the model-training process to keep the training consistent with the deployment. A combination of the predictive outputs of several models, Random Forest, Gradient boosting, and XGBoost are combined in the ensemble-inference module to generate a final classification. By using multiple classifiers instead of one the system benefits from the strength of collective decision-making on these and other matters, thus that becomes more robust, less variable and more stable in predicting. The ensemble module produces both the final predicted class and the corresponding confidence score which is used to create a meaningful risk assessment for the user.

The final architectural component is the output handler module which is responsible for formatting the model predictions into a human readable result. This module evaluates the input and assigns a level of risk for example, High, Medium, and Low using probability thresholds and message recommendations like “Avoid clicking this link” or “This URL seems trustworthy.” After that, the result is transmitted back to the mobile application’s Result Display Module which is visually rendered with colour-coded alerts and explanatory text.

In summary, the Design Phase establishes the entire structure of the system, functional components, flow logic and the layouts of the front end needed to success in iterative development of subsequent Agile cycles. The architecture supports modularity, reliability, and flexibility, all of which are important for a system that seeks to keep pace with ever-changing phishing threats.

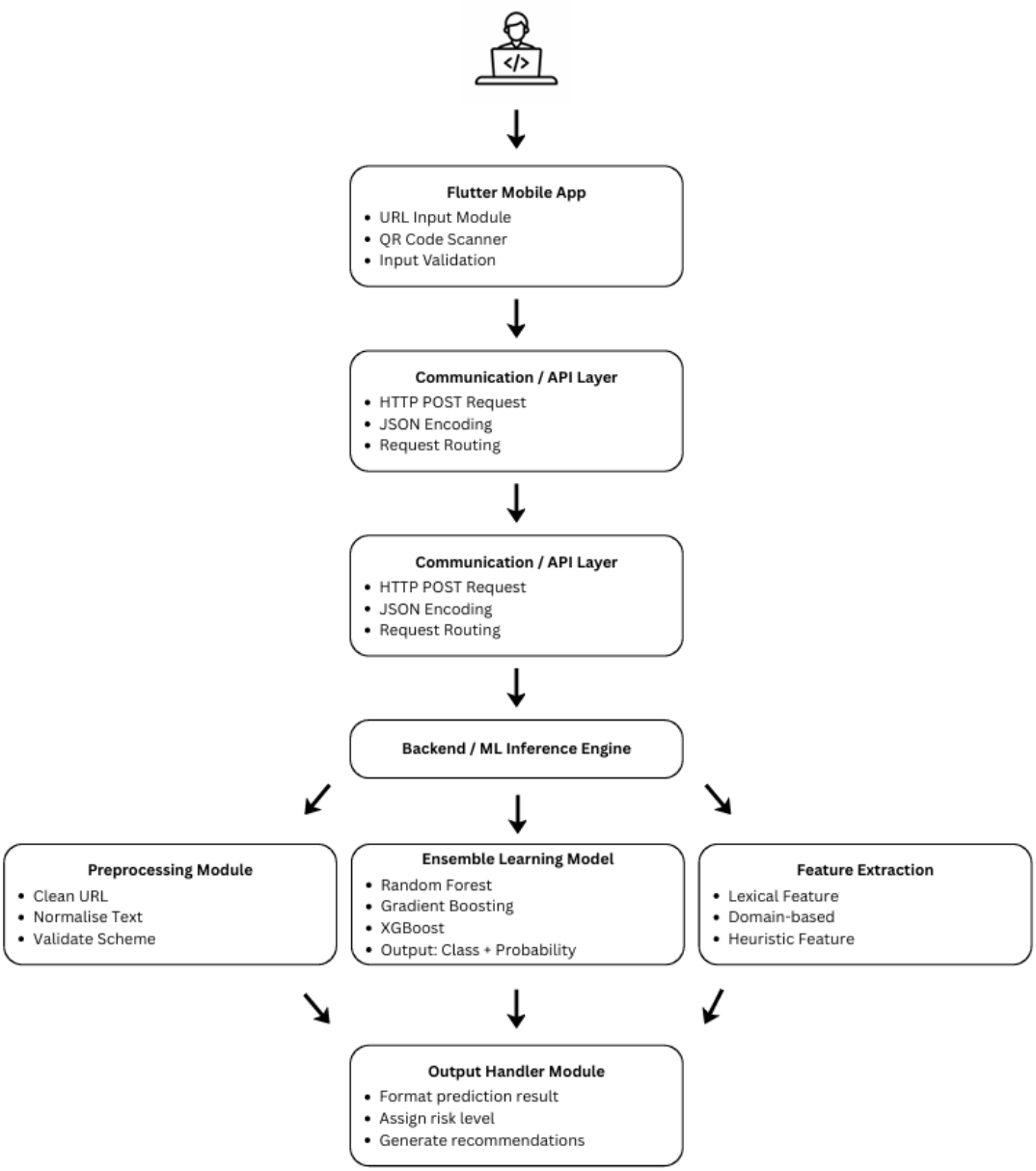

Figure 2: System Architecture of Phishing URL Detection System using Ensemble Learning

3.5 Development Phase

The Development Phase involves the build-up of the major components of the Phishing URL Detection System step by step using Agile cycles. Each cycle implements and test specific modules and improves them to ensure steady advancement in both machine learning pipeline and flutter mobile application.

The development starts with preparing the dataset and setting up the preprocessing workflow. The raw URLs are taken from Phish Tank, Open Phish, and Kaggle are cleaned to remove duplicates, invalid values, and formatting issues. Python scripts are designed to normalise the characters, validate the structures of the domains, and prepare the URLs to generate the features. This step ensures that any quality-related issues in the data are not existent throughout the whole machine learning pipeline.

After collecting the data, the feature engineering module converts each URL into its numerical equivalent ready to be used for training the model. Three categories of features like lexical, domain based and heuristic features are extracted with Python libraries such as re, tldextract and python-whois. These features capture patterns in the structures, domain age and ownership information and behavioural indicators concerning security. They are designed to be completely consistent with the features that were used during the training of the model to prevent discrepancies during development.

Table 4 : Summary of Key Feature Categories

Feature Category	Example Attributes	Purpose
Lexical Features	Length of URL, number of dots, presence of "@" and hyphens	Detect suspicious URL patterns and strange characters
Domain-Based Features	Check domain age, WHOIS register, and DNS status	Identify new or unusual domains that may be exploited for phishing
Heuristic Features	HTTPS flag, redirection count, and IP-based URL	Evaluate behavioural features related to phishing activities

Once feature engineering is finished, the next step is the development of the ensemble learning models. Random Forest, Gradient Boosting, and XGBoost are trained using the feature vectors prepared. Hyperparameters are optimised using several iterations to improve the model performance. After that the system combines the results of the three classifiers through some kind of ensemble mechanism such as, voting is one such approach, as is probability averaging, to provide a final and more reliable prediction. The ensemble model created is subsequently exported for further integration with the system's inference engine.

Development of the mobile application happens in parallel. Using the language Flutter, the main screens are built, which are the interface to enter the URL, the module to scan the QR code, and the result display page. Packages to access the camera and QR are built in to support the scanning function. Implementing the input validation and error handling mechanism to increase the user-friendliness of the application. Furthermore, the inference engine is then linked to the application via the application using HTTP requests (if backend is used) or on-device inference channels.

During this phase short Agile iterations are taking place that direct the development. Each module that is finished, such as the preprocessing, the feature extraction, the training of the model, the communication to the API, or the implementation of the UI (User Interface) is tested right away. Issues that are discovered during testing are corrected before moving on to the next cycle. This short, repetition cycle of development ensures that ML backend and the mobile application both evolve together and are in line with the overall system design.

3.6 Implementation Phase

The Implementation Phase is concerned with the integration of all the developed components into a functional and cohesive system. After the building of the machine learning models, feature extraction engine, communication layer and mobile interfaces, this phase ensures that each module works together seamlessly. Following the Agile methodology, the integration is done in iterations allowing for the early detection and correction of compatibility issues.

The first step of the implementation process is to deploy the ensemble learning model that has been trained. The model is either placed on a backend server or packaged for on-device inference, depending on the deployment strategy. In case of backend deployment, the ensemble model is placed in an inference service created with the help of such framework as Flask or Fast API. This service exposes an API endpoint that takes input of URL, preprocesses and extracts features, executes the ensemble inference mechanism, and returns a structured response in the form of a structured JSON response to the mobile application. For on-device deployment, the model is converted into a suitable format such as Tensor Flow Lite and incorporated into the Flutter app via platform channels, which allows inference without a network connection.

The next step is connecting the Flutter mobile interface up to the inference engine. API-based deployment involves the setup of secure (HTTPS) Post requests to send user submitted URLs or scanned QR data to the backend. The application is programmed to respond to the different kinds of responses, valid prediction or connection error or unexpected input. JSON parsing routines are implemented to

extract the classification results, scores of confidence and risk indicators from the output of the inference engine. If the system is based on local inference, there is no need for a network connection because the application calls the preprocessing and inference functions directly.

At the same time, the result display mechanism is also integrated to show the results of prediction in a clear and meaningful way. The classification result (whether phishing or legitimate), confidence score and the related risk level are presented through a colour coded indicator, for example, red showing for the high risk, yellow for moderate risk, and lastly green for low risk. Recommendation, text is also provided to help users understand the proper safety steps to take, such as being careful about suspicious links or being careful when interacting with unfamiliar domains.

In implementation Phase, numerous integration tests are conducted to ensure that the components communicate correctly with each other. The logic of preprocessing, feature extraction logic, and model inference operations are tested with an assortment of types of URLs, such as benign URLs, phishing URLs, malformed URLs and QR-encoded URLs. On the mobile side, the testing makes sure that the application properly transmits information, receives response, gracefully resolves errors and updates the user interface as well. Any anomalies found in the process of integration are resolved without delay before going to the subsequent repetitive process.

In general, the Implementation Phase helps to transform modules, which have been created separately, into a complete system. Through iterative Agile processes, the system becomes more stable and functional with each circle of integrating new elements into the system thus it is ready to be evaluated, and performance tested during the next stage.

3.7 Evaluation Phase

In Evaluation Phase, it is very important in determining the performance and trustworthiness of the Phishing URL Detection System. Moreover, this stage guarantees that the mobile application and the ensemble learning model are functioning correctly together after integration. Evaluation is carried out in an iterative manner so that defects can be discovered and improvements made.

The evaluation model is evaluated based on accuracy, precision, recall, F1-score, and confusion matrix. These metrics are crucial because phishing detection requires a detection rate practically equal to the number of malicious URLs while allowing as few legitimate URLs to be flagged as false positive as possible. The final ensemble model is investigated with a dedicated test corpus to validate the generalisation capabilities of the model to previously unseen URLs. Any inconsistencies in the model's predictions such as lower accuracy or fluctuations in predictions are addressed using hyperparameters tuning and feature selection.

After the model evaluation, the mobile application conducts a system level evaluation. Functional tests are the one that ensure the input of URLs, the scanning of QR codes, the communication with the backend and the presentation of results work without interruption. Integration tests check if the application communicates properly with the inference engine and handles errors correctly and usability tests check if the user can navigate through the interface without any problems and understand the risk levels without ambiguity.

Table 5 : Functional Testing Summary

Test Objective	Description	Expected Outcome
URL Input Validation	Testing of valid, invalid, and malformed URLs	App accepts valid URLs and rejects invalid formats
QR Code Scanning	Scanned URLs from printed and screen-based QR codes	QR code is read accurately and converted to a URL
Backend / Model Integration	Send URL to inference engine and receive prediction	System returns classification and confidence correctly
Output Display	Readability of results and risk indicators is evaluated	User sees clear classification, colour indicators and advice

Through these evaluative procedures, both the machine learning backend and the mobile interface are improved to ensure stability, precision and ease of user interaction. The effectiveness of these processes is an indication of the system being ready to enter the last development and testing phases.

3.8 Summary and Toolsets for the Project

This part of the project offers an overview of the tools and technologies that have been chosen to support the design activities in FYP 1 are presented in this section of the project. Since the actual system development will be done in FYP 2, the toolsets presented here are rather for exploring the datasets, formulating methodology, and modelling the architecture. Additionally, these tools will provide the technological backbone for the implementation of the system during the next phase, so that there is consistency between the conceptual design in FYP 1 and the practical development in FYP 2. By specifying the tool environment from the very beginning, the project will be clear and prepared on a solid basic for the upcoming ensemble learning model, mobile application and back-end inference engine.

Table 6 : Tools and Technologies Summary for Phishing URL Detection System using Ensemble Learning

Category	Tool and Technology	Objective
Programming and Machine Learning Tools	Python	Main language for data preprocessing and ensemble learning model development
	Scikit-learn	Offers Random Forest, Gradient Boosting, and preprocessing tools
	XGBoost Library	Allows for the training and evaluation of the XGBoost component in the ensemble
	Pandas & NumPy	Applied for managing data sets, numerical computation, and feature engineering
Dataset Source	Phish Tank	Supplied verified phishing URLs for research and model preparation

	Open Phish	Provides real-world phishing URLs to increase the diversity of the dataset
	Kaggle Datasets	Used to collect the legitimate URLs to balance and strengthen the dataset
Backend and Deployment Tools	Flask / Fast API	Frameworks for building the backend API that connects the mobile app the ML inference engine
	Tensor Flow Lite	Used to convert models into lightweight format for on-device deployment in Flutter
Mobile Development Tools	Flutter	Framework for developing the mobile application that works on multiple platforms
	Dart	Programming language for the mobile app logic and UI components
	Flutter QR Scanner Plugin	Enables QR code scanning to extract URLs within the mobile app
Design and Diagramming Tools	Canva	Used to design the system architecture diagram
	Figma	For creating UI mock-ups and planning user interface layouts

In summary, the current chapter has set the methodology foundation for Phishing URL Detection System using ensemble learning. A hybrid SDLC approach was selected to combine structured planning with iterative refinement and hance suitable for research-driven FYP 1. The chapter outlined the general workflow of the proposed machine learning pipeline that includes planning of preprocessing, feature engineering approaches, ensemble model design for the pipeline, and system architecture for the backend and mobile application. As FYP 1 is only focused on documentation, therefore the focus was guided towards conceptual preparation rather than implementation. Furthermore, the toolsets mentioned in this part are applicable to all research, design, and planning activities while their importance will still be high in the development phase of FYP 2. With a solid methodology and all-encompassing design framework, the project is now ready for implementation in the next trimester.

CHAPTER 4

IMPLEMENTATION AND EVALUATION

4.1 Overview of System Implementation

In this chapter, the implementation and evaluation of the proposed Phishing URL Detecting System using Ensemble Learning is presented. Unlike Chapter 3 that seeks to focus on the planned methodology, system design and development workflow, this chapter focuses on the practical realisation and validation of the proposed approach at the Final Year Project Phase 1 (FYP 1) level.

Chapter 3 explains what methods and phases are intended to be used while the details of how these methods are applied in practice through prototype implementation and structured evaluation are given in Chapter 4. As FYP 1 is more focused on conceptual validation instead of full-scale deployment, implementation discussed in this chapter represents a functional prototype more for feasibility demonstration, system integration testing and detection effectiveness, not a production ready system.

The implementation phase is focused on converting the conceptual phase into a working prototype, which can analyse URLs and QR codes to identify phishing threats. The system is based on the design described in Chapter 3, combining the mechanisms for feature extraction, ensemble learning classifiers, and a mobile interface developed with Flutter technologies. The evaluation phase then verifies that the system fulfils the project objectives through the analysis of model accuracy, robustness and usability.

4.2 Implementation of the Phishing URL Detection System

4.2.1 Dataset Preparation and Preprocessing

The implementation starts with dataset preparation, which is a very important step to ensure the reliable training and evaluation of the model. Therefore, the implementation starts with this process. Sets of phishing and benign URLs that are

publicity available were sourced from different places like Phish Tank, Open Phish, and Kaggle. Combining these datasets helped to improve the diversity, thus removing the bias of a certain phishing pattern.

Preprocessing was carried out prior to model training to clean and standardise the data. The process involved removing duplicate URLs, supplying missing values, and filtering out malformed URLs. All the URLs were converted into one format so that the features could extracted accurately. This preprocessing step helps in reducing noise and avoiding inconsistencies that can have a negative impact on the ensemble models.

4.2.2 Feature Extraction Implementation

After some preprocessing, each URL is converted into a numerical feature vector using some structured process of feature extraction. The implementation stresses lightweight yet effective features of the URL to assure that the system is suitable for the mobile implementation without excessive computational overhead. The features that have been extracted are intended to capture the distinguishing characteristics between the legitimate and phishing URLs and yet be efficient and scalable.

The extraction of features is conducted into three major categories which is lexical, domain based, and heuristic. Lexical features directly come from the textual structure of the URL and contain features such as length of URL, number of dots, presence of special characters, IP addresses instead of domain names. Furthermore, domain-based features are extracted to present contextual information regarding the URL, such as the domain age and registration status, which are usually related to phishing behaviour. Heuristic features are also added to reflect security-related features, such as the existence of HTTPS and possible redirection patterns.

These features are extracted from using python-based scripts and supporting libraries such as regular expressions and domain parsing tools. This is because to ensure that the results are reliable and consistent, the same feature extraction process

is applied in both training and testing stages of the model. This consistency guarantees that the ensemble learning models are assessed under the same circumstances as they are trained, which can help to improve the validity of the evaluation results.

4.2.3 Ensemble Learning Model Implementation

In this system, three ensemble learning algorithms were implemented which are Random Forest, Gradient Boosting, and XGBoost. The extracted feature vectors were separately trained to each model. Hyperparameters were tuned using an iterative experimentation method to achieve the best performance and to minimize the risk of overfitting.

The ultimate ensemble choice was made with a probability-based voting scheme. Instead of using a single classifier, ensemble combines the prediction probabilities from all three classifiers and produces a final classification depending on an aggregated score of confidences. This approach helps in better robustness and less chance of misclassification, especially for the previously unseen phishing URLs.

4.2.4 Mobile Application Implementation

The mobile application is developed with the Flutter framework so that it can be deployed on both Android and iOS mobile devices. The application is designed to offer a simple and intuitive interface to the users for phishing detection and to retain compatibility across various mobile platforms. This is done by providing the users with two major modes of detection included manual URL input and QR code scanning.

Through the manual URL input feature, users can paste or type a URL into the application to be analysed. The QR code scanning feature is used by the user to detect phishing links, which are embedded within QR codes, using the device's camera, which is especially useful when trying to identify threats from printed materials or digital displays. Input validation mechanisms are implemented to ensure that only properly formatted URLs are processes by the system which helps in reducing the errors and increasing the reliability of the system.

Once the URL is submitted, it is sent to the inference engine for classification, and the detection result is returned to the application. The result page displays the classification result with the help of colour-coded indicators for better understanding of the user. Phishing URLs are shown in red and legitimate URLs are shown in green. In addition, a short explanation and safety recommendation is provided to help users to interpret the result and take appropriate action. This design approach focuses on clarity, ease of use and accessibility, making the application usable by people with different levels of technical knowledge.

4.3 Implementation of the Phishing URL Detection System

This section explains the evaluation method taken to test the feasibility and effectiveness of the proposed phishing URL detection system in the Final Year Project Phase 1 (FYP 1) stage. The evaluation is done at a prototype level to validate the correctness of the implementation and the appropriateness of the chosen ensemble learning approach and not to make any final or production-level performance claim. The assessment is focused on the capability of the system to perform phishing detection reliably when tested in controlled testing conditions.

Moreover, the evaluation process involves testing the trained ensemble model on a reserved dataset which not used in the training process. In this way, the evaluation process will be able to reflect the ability of the model to generalise to unseen data. The preprocessing and feature extraction steps of the training data are the same as those used in the above sections and are used uniformly in the evaluation process to ensure reliability and fairness of performance measurement.

To measure the performance of the proposed system some standard classification metrics are employed. The metrics that are used include accuracy, precision, recall, and F1-score which are commonly used in phishing detection and binary classification tasks. Accuracy indicates the overall correctness of the model, while precision shows how many of the URLs that have been labelled as phishing are harmful. Recall measures the model's ability to identify phishing URLs, and F1-score gives a balance that takes into consideration of both precision and recall. For clarity

and ease of interpretation of the information, all metric values are presented in percentage format.

4.4 Evaluation Metrics and Performance Criteria

Since this project is being undertaken in Final Year Project Phase 1 (FYP 1), the evaluation aspects are placed on defining appropriate performance metrics which will be used to evaluate the effectiveness of the proposed ensemble learning model in Phase 2. At this point, the machine learning model has not yet been completed in implementation and training. Thus, no numerical performance scores are available at this stage. Instead, this section describes evaluation measures planned to be used in FYP Phase 2.

Table 7 : Evaluation Metrics for Phishing Detection System

Metric	Description	Importance in Phishing Detection
Accuracy	Measure the overall correctness of the classification by calculating the percentage of URLs assigned to the right category, compromising both phishing and legit URLs.	Provides a general overview of the model's performances but could be misleading if datasets are imbalanced as is the case with phishing detection.
Precision	Measure the proportion of URLs ranked as phishing are in fact phishing.	Helps to minimize false positives by making sure that legitimate URLs are not flagged falsely which will increase user trust in the system.
Recall	Determines the percentage of actual phishing URLs that the system has correctly identified.	Important for security applications, as the failure to detect phishing URLs can put users at a high risk.

F1-score	The harmonic average of the precision and recall, so that both the false positive and false negative are balanced.	Provides a balanced evaluation of the effectiveness of the system in case where there is a trade off between precision and recall.

These evaluation metrics are popular in phishing detection studies and provide a comprehensive measure of the classification performance. in FYP Phase 2, these metrics will be used to analyse experimental results that are retrieved from trained models and testing datasets to quantitatively assess the effectiveness of the proposed ensemble learning-based phishing detection system.

4.5 System and Usability Evaluation

In addition to assessing the performance of the models, a system-level testing was performed to check the overall functionality and usability of the phishing URL detection system. The verification is primarily concerned with making sure that the integrated system's components work well together and that the user experience is good enough to meet the standards set for the practical implementation of the system in a mobile application. The evaluation at the system level was done using a prototype stage that was defined by the scope of the Final Year Project Phase 1.

Functional testing was conducted to ensure that the main functions of the system are correct. The results show that the system can provide both manual URL input and QR code scanning without errors. URLs that are entered by manual users are successfully checked and sent to the inference engine for classification and URLs that are extracted from QR codes are decoded and analysed correctly. Integration testing further ensured that communication between the interface of the mobile application and the backend inference engine is smooth, so classification results are returned and shown without visible delay or failure.

Usability testing was concentrated on testing the clarity of the mobile application interface, the responsiveness, and usability. The application layout, colour-

coded presentation of the results and concise safety recommendations all contribute to making the user experience intuitive. The evaluation shows that the users can easily interpret the results of the classification and understand the suggested actions without having technical expertise. This usability focus is important for phishing detection applications because these applications are designed to be used by a wide range of different users. Overall, the results of the system-level and usability evaluation indicate that the proposed system satisfies the functional goals and offers a viable basis for further improvement and user-based testing for Final Year Project Phase 2.

4.6 Summary

This chapter introduced the implementation and the initial evaluation of the phishing URL detection system developed for Final Year Project Phase 1. The chapter started with the explanation of the system implementation that included data preprocessing, feature extraction, ensemble model integration, and mobile application development using the Flutter framework. To assure mobile deployment, lightweight yet powerful URL-based features have been applied while the ensemble learning method have been applied while the ensemble learning method has been adopted to enhance detection reliability. The mobile application has been designed keeping the accessibility to users in mind with manual URL input and QR code scanning functionalities and clear, colour-coded detection feedback. Following the implementation discussion, the evaluation methodology and performance metrics were presented with the aim of evaluating the feasibility of the proposed system at a prototype level. Standard classification metrics, such as accuracy, precision, recall and F1-score were employed to validate the effectiveness of the ensemble model on a reserved test dataset. Evaluation results shows that the proposed approach is able to distinguish phishing URLs from legitimate URLs in a controlled environment and thus demonstrate the feasibility of the approach to be refined and fully evaluated in Final Year Project Phase 2.

CHAPTER 5

CONCLUSION

5.1 Conclusion and Future Work

This chapter concludes the Final Year Project Phase 1 (FYP 1) by summarising the work achieved, reflecting on the achievement of the project stated objectives, highlighting the implications of key insights gained during the planning and design stages and providing an indication of future work to be carried out in Final Year Project Phase 2 (FYP 2).

5.2 Summary of Achievements from the Project

The main goal of this project is to design Phishing URL Detection System using Ensemble Learning techniques where the focus is more on mobile-based detection and QR code analysis. During FYP Phase 1, the project has been able to develop a strong conceptual and methodological basis needed for system implementation in the following phase.

Besides, a detailed literature review was performed to investigate the current phishing detection method, ensemble learning methods, mobile-based security solutions, and the new threat of QR code phishing. Through critical analysis, a numerous of research gaps emerged, notably, the lack of lightweight, mobile-friendly phishing detection systems with an effective integration of ensemble learning in QR code scanning capabilities.

In addition, the hybrid model of Software Development Life Cycle (SDLC) was chosen and justified as a balance between structured planning and iterative refinement. The overall system architecture, machine learning pipeline, and mobile application workflow were well thought out to achieve the analysis of URL and phishing classification. Moreover, suitable datasets, feature extraction approaches, or ensemble classifiers and development tools too were identified and documented, to ensure that they were ready to be implemented.

5.3 Achievement of Project Objectives

The objectives defined in Chapter 1 have been tackled at Final Year Project Phase 1 level by the activities which conducted systematically in research, planning, and design. The first objective, which concerns the study of the techniques of phishing and the existing means of detection, was accomplished with an extensive literature review and critical study of existing detection methods of phishing. This analysis gave some idea of modern ways of phishing, current limitations, and the research gaps motivating the proposed solution.

The second objective which aims to design an ensemble learning-based phishing URL detection model was achieved through the conceptual design of the machine learning pipeline that contains multiple classifiers, namely Random Forest, Gradient Boosting, and XGBoost. The selection of these models and how they were integrated was meticulously planned to increase the detection accuracy and strength of models especially in the recognition of complex phishing patterns.

The third objective, which includes the design of a mobile application architecture with both URL and QR code scan capabilities, was achieved through the design of a detailed system architecture and mobile application design in the form of a mobile application using the Flutter technology. This design supports cross-platform compatibility as well as the functionality needed by phishing attacks that are of the most importance in real world detection on mobile devices.

Finally, the goal to plan suitable evaluation metrics for system performance was accomplished by identifying common performance measures such as accuracy, precision, recall, and F1-score. These metrics will serve their purpose in FYP Phase 2 to evaluate the proposed system's efficiency and reliability. Although full system implementation and experiment testing are not within the scope of FYP Phase 1, all the objectives related to planning, design and feasibility analysis have been satisfactorily accomplished.

5.4 Limitations of FYP Phase 1

The limitations are inevitable since FYP Phase 1 is primarily focused on documentation, planning, and conceptual validation. The proposed system still needs to be completely implemented or deployed, and no real-time testing has been done. In addition, advanced features such as real-time threat intelligence integration, large-scale deployment and deep webpage content analysis were intentionally excluded to ensure the project is feasible within academic constraints.

These limitations are suitable for FYP 1 and will be solved incrementally in FYP Phase 2.

5.5 Future Work (FYP Phase 2)

In FYP Phase 2, the project will progress from the conceptual design to full implementation and evaluation. The ensemble learning models will be trained and test with real-world datasets of phishing and valid URLs. The features extraction and the application of preprocessing methods will be adjusted for the purpose of boosting the classification accuracy and strength.

Furthermore, the mobile application will be developed completely under the framework of Flutter and will have both manual URL input and QR code scanning functionality. Performance evaluation, usability testing and optimisation of the systems will also be done. Potential improvements can be added such as the optimisation of models for on-device inference, better user interface design, and the addition of other security features to increase user awareness.

5.6 Chapter Conclusion

In conclusion, FYP Phase 1 has successfully established a good and structured foundation for the development of a Phishing URL Detection System using Ensemble Learning. The project is well-placed to move forward to the implementation and evaluation phases in FYP Phase 2. The results of this phase demonstrate the feasibility, relevance, and academic worth of the Phishing URL Detection System using Ensemble Learning in tackling modern phishing threats.

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APPENDICES

Appendix A: Meeting Log

Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

(To be filled in by Student)

MEETING DATE: 7/11/2025	**MEETING NO.: 1**
PROJECT ID: T88J252	
PROJECT TITLE : PHISHING URL DETECTION SYSTEM USING ENSEMBLE LEARNING	
TRIMESTER : Oct/Nov 2025 (2530)	**SUPERVISOR : Alvin Lim Fang Chuen**
STUDENT ID & Name: 1211101093 Eddie Chin Jia Ze	**CO- SUPERVISOR :**

1. **WORK DONE** [Please write the details of the work done after the last meeting.]

- I have started reading and gathering information about phishing detection and ensemble learning from research papers and online sources.
- I have explored the tools that I might need to use such as Kaggle and Virtual Studio Code.

2. **WORK TO BE DONE**

- Start writing the first chapter of the Final year project (FYP) report.
- Plan how to carry out the project, including how to collect data and build detection system.
- Find a good dataset of URLs and begin to clean and analyse the data.

3. **PROBLEMS ENCOUNTERED**

- Developing the QR code scanning feature is difficult for me since I have never built one before. I am unsure how to begin integrating it with the Phishing URL Detection System.

4. **COMMENTS**

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Malaysia.

Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)
**: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

(To be filled in by Student)

MEETING DATE: 21/11/2025	**MEETING NO.: 2**
PROJECT ID: T88J252	
PROJECT TITLE : PHISHING URL DETECTION SYSTEM USING ENSEMBLE LEARNING	
TRIMESTER : Oct/Nov 2025 (2530)	**SUPERVISOR : Alvin Lim Fang Chuen**
STUDENT ID & Name: 1211101093 Eddie Chin Jia Ze	**CO- SUPERVISOR :**

1. **WORK DONE** [Please write the details of the work done after the last meeting.]

- I have started writing Chapter 1 , including the Introduction, Background of the study, and Problem Statement.
- I refined my understanding on phishing attacks and potential use of ensemble learning for phishing URLs detection by reading more articles and examples related to it.

2. **WORK TO BE DONE**

- Continue to improve Chapter 1 by writing the project objectives, scope, and significance of the study.
- Start searching for academic journals and conference papers for preparing the chapter of literature review.

3. **PROBLEMS ENCOUNTERED**

4. **COMMENTS**

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Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

(To be filled in by Student)

MEETING DATE: 28/11/2025	**MEETING NO.: 3**
PROJECT ID: T88J252	
PROJECT TITLE : PHISHING URL DETECTION SYSTEM USING ENSEMBLE LEARNING	
TRIMESTER : Oct/Nov 2025 (2530)	**SUPERVISOR : Alvin Lim Fang Chuen**
STUDENT ID & Name: 1211101093 Eddie Chin Jia Ze	**CO- SUPERVISOR :**

1. **WORK DONE** [Please write the details of the work done after the last meeting.]

- I have completed most parts of Chapter 1, including objectives, project scope, and report structure.
- I began collecting research papers related to phishing detection techniques, URL-based features, QR code phishing, and ensemble learning models.

2. **WORK TO BE DONE**

- Revise Chapter 1 after received supervisor feedback.
- Start writing Chapter 2 by summarizing and comparing related studies found in the literature.

3. **PROBLEMS ENCOUNTERED**

- Some research papers are hard to understand because of the technical terms and complicated explanations.

4. **COMMENTS**

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Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

NOTES:

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7. For FYP Phase 2, total six log sheets are to be submitted (every other week**).

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Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

(To be filled in by Student)

MEETING DATE: 5/12/2025	**MEETING NO.: 4**
PROJECT ID: T88J252	
PROJECT TITLE : PHISHING URL DETECTION SYSTEM USING ENSEMBLE LEARNING	
TRIMESTER : Oct/Nov 2025 (2530)	**SUPERVISOR : Alvin Lim Fang Chuen**
STUDENT ID & Name: 1211101093 Eddie Chin Jia Ze	**CO- SUPERVISOR :**

1. **WORK DONE** [Please write the details of the work done after the last meeting.]

- I had done the first draft of Chapter 2 Literature Review by summarizing the previous research related to phishing detection and machine learning methods. Compared the different approaches used by other researchers and identified common techniques and limitations.

2. **WORK TO BE DONE**

- Continue Chapter 2 by adding more discussion of research gaps and how my project can address these.
- Make sure all the references are cited appropriately.

3. **PROBLEMS ENCOUNTERED**

4. **COMMENTS**

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Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

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7. For FYP Phase 2, total six log sheets are to be submitted (every other week**).

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Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

(To be filled in by Student)

MEETING DATE: 19/12/2025	**MEETING NO.: 5**
PROJECT ID: T88J252	
PROJECT TITLE : PHISHING URL DETECTION SYSTEM USING ENSEMBLE LEARNING	
TRIMESTER : Oct/Nov 2025 (2530)	**SUPERVISOR : Alvin Lim Fang Chuen**
STUDENT ID & Name: 1211101093 Eddie Chin Jia Ze	**CO- SUPERVISOR :**

1. **WORK DONE** [Please write the details of the work done after the last meeting.]

- I have completed chapter 2 and improved the discussion of ensemble learning models like Random Forest, Gradient Boosting, and XGBoost.
- Discussed the proposed system approach and overall project direction with my supervisor.

2. **WORK TO BE DONE**

- Start for Chapter 3, focusing on the project methodology, system architecture, and development approach.
- Decide for the tools and the technologies to be used in the system development.

3. **PROBLEMS ENCOUNTERED**

4. **COMMENTS**

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Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

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7. For FYP Phase 2, total six log sheets are to be submitted (every other week**).

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**: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

(To be filled in by Student)

MEETING DATE: 26/12/2025	**MEETING NO.: 6**
PROJECT ID: T88J252	
PROJECT TITLE : PHISHING URL DETECTION SYSTEM USING ENSEMBLE LEARNING	
TRIMESTER : Oct/Nov 2025 (2530)	**SUPERVISOR : Alvin Lim Fang Chuen**
STUDENT ID & Name: 1211101093 Eddie Chin Jia Ze	**CO- SUPERVISOR :**

1. **WORK DONE** [Please write the details of the work done after the last meeting.]

- I have completed chapter 3, including system workflow, development methodology, and system design.
- I have discussed how phishing URLs are being processed and how ensemble learning models will be applied in the detection system.

2. **WORK TO BE DONE**

- Preparing for Chapter 4 by planning the implementation details and evaluation method.
- Review on previous chapters to make the report consistent.

3. **PROBLEMS ENCOUNTERED**

4. **COMMENTS**

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Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

(To be filled in by Student)

MEETING DATE: 9/12/2025	**MEETING NO.: 7**
PROJECT ID: T88J252	
PROJECT TITLE : PHISHING URL DETECTION SYSTEM USING ENSEMBLE LEARNING	
TRIMESTER : Oct/Nov 2025 (2530)	**SUPERVISOR : Alvin Lim Fang Chuen**
STUDENT ID & Name: 1211101093 Eddie Chin Jia Ze	**CO- SUPERVISOR :**

1. **WORK DONE** [Please write the details of the work done after the last meeting.]

- I have completed chapter 4 with the explanation of system implementation and basic evaluation of the proposed phishing detection system.
- I started writing Chapter 5 by summarizing the overall project, objectives achieved, and system limitations.

2. **WORK TO BE DONE**

- Complete the Chapter 5 and add future improvement suggestions.
- Check the entire report, such as formatting, grammar, and consistency before submit the report.

3. **PROBLEMS ENCOUNTERED**

4. **COMMENTS**

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Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

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Appendix B: LLC Attendance

FYP for FIST: Attendance List
Session 1: 26/11/2025

The following students have attended the FYP session (FIST) with the library.

Trainer:

MELISSA DE VALDA BINTI MOHD YATIM
Assistant Manager, Library
Information Skill & Research Support Unit
Multimedia University
Jalan Ayer Keroh Lama
75450 Melaka

Melissa De Valda Binti Mohd Yatim
Librarian, Tun Dr. Hasmah Mohd Ali Digital Library

69	Priscilla Goh Yi Suen	1211106676	Phishing and Malicious Url Detection using AI
70	Ho Yi En	241UT2410J	Computer-Vision Based Object Detection Models in Traffic-based Scenarios
71	LEONG YU HEN	242UT244BU	MOBILE-BASED TAPE MEASUREMENT APPLICATION USING CAMERA AN
72	WONG BING YUAN	1211107102	SMART CAREER PREDICTION AND GUIDANCE SYSTEM
73	PHUNG YI XUAN	1211109738	AI SALES MANAGEMENT SYSTEM FOR MICRO VENDORS
74	eddie chin jia ze	1211101093	Phishing URL Detection System using Ensemble Learning
75	Foong Han Le	1211104983	Secure File Storage System using Encryption
76	LIM MING QUAN	243UT246XS	Next-Gen E-Learning: Implementing AI-Driven Adaptation in Digital Education
77	Ang Zhen Loong	1211102829	smart monitoring and performance analytics for postgraduate students
78	CHONG TIAN WEI	1231303630	STACKING-BASED HEART DISEASE CLASSIFICATION MODEL
79	HEE WEI ZHE	242UT24499	STUDENT COUNTING SYSTEM USING MOBILE CAMERA FOR SECURE RE
80	LEE XIN ZHE	242UT24497	DEVELOPMENT OF A FINGER VEIN RECOGNITION SYSTEM BASED ON T
81	LOH PEI PEI	1221102844	ENHANCED SALES FORECASTING THROUGH MACHINE LEARNING
82	CHIA ZI QI	1211107236	Vision-based food tracking system
83	Ding Teck Sheng	1211110323	Zero Emission Building: Automated Carbon Footprint Auditing By IOT Edge
84	ONG HAI MEI	242UT242YG	CONTINUAL LEARNING FRAMEWORK FOR IMPROVING LARGE LANGUAG
85	Chong Yong Khong	1171201770	Implementation of Deep Learning for Text Classification on Public Sentiment

Appendix C: Checklist for FYP Interim Submission

Faculty of Information Science and Technology (FIST)

Checklist for Interim Report Submission

(To be filled in by Student)

STUDENT'S DETAILS

Project Code	T88J252
Name	Eddie Chin Jia Ze
ID No	1211101093
Title of Thesis	Phishing URL Detection System using Ensemble Learning
Supervisor Name	Alvin Lim Fang Chuen

REPORT ARRANGEMENT	√	**Comments** **(if any differences)**
1. Cover of The Interim Report	√	
2. Title Page of the Interim Report	√	
3 Copyright page of I Interim Report	√	
4. Declaration Page of Interim report	√	
5. Acknowledgement	√	
6. Table of Contents	√	
7. Abstract	√	
8. List of Tables	√	
9. List of Figures	√	
10. List of Symbols	√	
11. List of Appendices	√	
12. Chapter 1: Introduction – objectives, scope	√	
13. Chapter 2: Literature Review	√	
14. Chapter 3: Title: Methodology	√	
15. Chapter 4: Title: Implementation & Evaluation	√	
16. Chapter 5: Title: Conclusion	√	
17. References – APA style	√	
18. Appendices	√	
19. CD/ DVD and envelope as shown in Appendix K		No require in this project
20. Attachment : FYP Meeting Logs (all) 1 set	√	

FORMAT OF REPORT	√	Comments
1. Page Numbering	√	
2. Font and Type Face	√	
3. Font Cover	√	
4. Tables and Figures	√	
5. Comb Bind		No hardcopy
6. Colour of the Front Cover		No hardcopy
7. Number of words > 5000 (Main content only)	√	

Checked by

21/1/2026 1:05 p.m

Student's Signature & Date